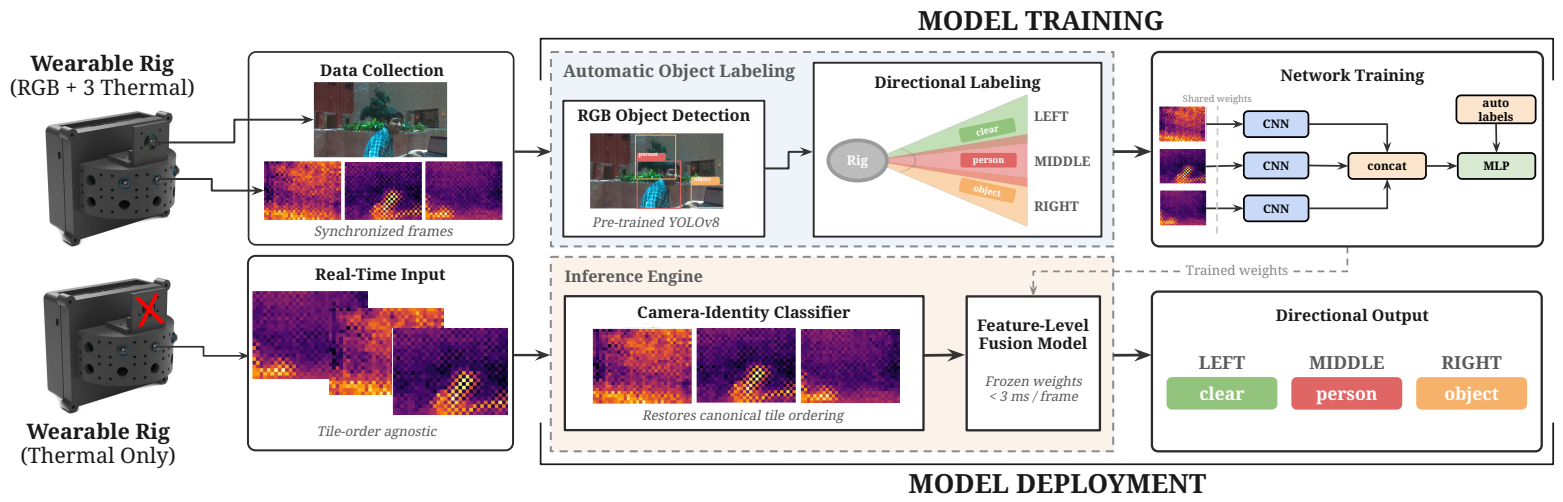

MODEL TRAINING
Wearable Rig
(RGB + 3 Thermal)
Data Collection
Synchronized frames
Automatic Object Labeling
RGB Object Detection
Pre-trained YOLOv8
Directional Labeling
Rig
clear
person
object
LEFT
MIDDLE
RIGHT
Network Training
Shared weights
CNN
CNN
CNN
concat
MLP
auto labels
Trained weights
Wearable Rig
(Thermal Only)
Real-Time Input
Tile-order agnostic
Inference Engine
Camera-Identity Classifier
Restores canonical tile ordering
Feature-Level Fusion Model
Frozen weights
< 3 ms / frame
Directional Output
LEFT
MIDDLE
RIGHT
clear
person
object
MODEL DEPLOYMENT